

a Evidence register and permitted claim scope

Figure 4 integrates earlier analyses; rows are not independent replications and do not form a validation ladder.

STATUS	EVIDENCE DOMAIN	OBSERVED SUMMARY	PERMITTED INTERPRETATION
DESCRIPTIVE	Family recurrence	248 modes · 43 datasets · 9 contexts 16 unsupervised families	Descriptive, run-relative structure; not a transportability estimate
CONDITIONAL	Same-label residual transfer	Leave-dataset cosine 0.563 study-proxy exclusion 0.035	Association inside the assembled pipeline; broad transport is not established
PREDICTIVE	Shared-endpoint fitness prediction	Mean fold ρ 0.622 (5 folds) 1,229-gene benchmark	Predictive for the frozen gene benchmark; grouped holdouts are unstable
POST HOC	Curated pathway coherence	9 displayed candidate annotations study share 0.643–1.000	Post hoc, study-conditioned shorthand; not independent biological replication
NOT TESTED	Causal mechanism	No prospective intervention or external-study mechanism test	Universal axes, mechanism, and causal law are not established

b Study-conditioned candidate summaries

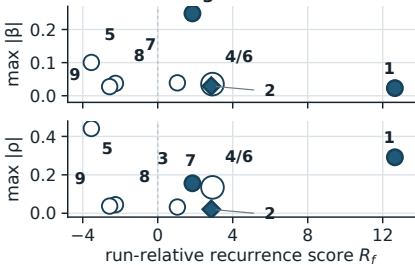
Raw maxima from revised Figure 3; shading is scaled separately within each column and does not define a combined score.

#	CANDIDATE LABEL	TIER*	RECURRENT R_f	same residual coordinates		same DepMap endpoint		
				ORA $-\log_{10}(q)$	RANK $-\log_{10}(q)$	MAX $ \beta $	MAX $ \rho $	MAX STUDY SHARE
1	Cytokine / JAK-STAT	H	12.65	54.7	16.0	0.023	0.291	1.000
2	Chromatin / epigenetic	H	2.86	14.5	NR	0.029	0.020	0.875
3	Translation / RQC	H	1.85	9.9	21.1	0.249	0.156	1.000
4	Chromatin-cytokine	S	2.92	39.0	21.5	0.035	0.135	1.000
5	OXPPOS / TCA	S	-3.54	8.9	11.1	0.101	0.441	1.000
6	Lysosome / ECM	S	2.92	5.3	4.3	0.035	0.135	1.000
7	rRNA biogenesis	S	1.05	4.3	3.3	0.039	0.032	1.000
8	SREBP / cholesterol	S	-2.25	6.1	5.1	0.037	0.044	1.000
9	Antigen-receptor / tolerance	S	-2.56	3.1	3.6	0.028	0.037	0.643

*Manual tier: H, headline; S, supporting. NR, no retained rank result—not zero. Metrics are dependent; unlike units are not commensurate.

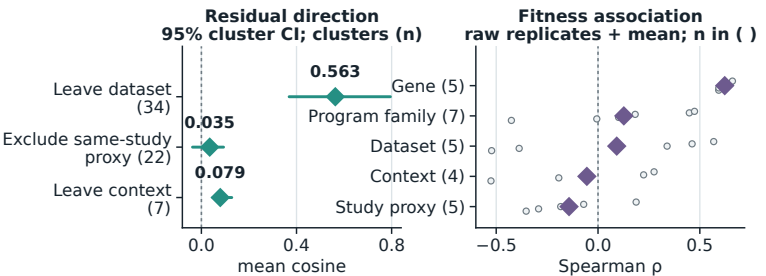
c Raw-metric atlas

No composite, area, or jitter.
4/6 coincide; 2 overlaps only in β .
 β and ρ reuse one endpoint.

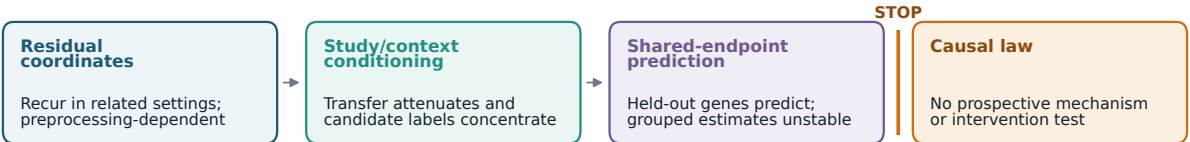


d Generalization boundaries

Different estimands and resampling units;
tracks are not quantitatively pooled.



e Scoped model and decisive falsifiers



Required for broader claims: fold-fitted preprocessing · verified study IDs · independent-study recurrence
declared contrast basis · study-block uncertainty · prospective mechanism/intervention tests